Clearance & Elimination Practice 3

Module 4 - Drug Elimination and Clearance · Daily practice sheet · No date

Name _____ Date _____

Cefprozil, 500 mg oral every 12 hours for 10 days, has been prescribed for patient, KL. KL is a 42 year old male, who is 178 cm tall and weighs 73 kg and has a serum creatinine of 0.82 mg/dL. Cefprozil is well absorbed with a bioavailability of about 95%. The half-life of elimination in adults with normal hepatic and renal function is about 1.3 hours, 5.2 hours in adults with renal impairment (120 > CrCl > 30mL/min), and 5.9 hours in patients with renal failure (CrCl< 30mL/min),. The apparent volume of distribution is 0.23 L/kg. About 60% of the drug is excreted in the urine as unchanged drug. According to the manufacturer's labeling for adults, no dosage adjustment is necessary if the CrCl ≥30 mL/minute and the dose should be reduced by 50% if CrCl <30 mL/minute.

a Estimate KL’s creatinine clearance.

b Estimate the total body clearance (mL/min) of cefprozil in KL.

c Estimate the renal clearance (mL/min) of cefprozil in KL.

d Estimate the hepatic clearance (mL/min) of cefprozil in KL.

e If KL developed some severe complications and his serum creatinine sky-rocketed to 2.8 mg/dL, revise your estimate of KL's creatinine clearance.

f Provide a new estimate of the total body clearance (mL/min) of cefprozil in KL assuming his volume of distribution is unchanged.

g Provide a new estimate of the renal clearance (mL/min) of cefprozil in KL.

h Provide a new estimate of the hepatic clearance (mL/min) of cefprozil in KL.

Solutions

Her key: the answer and working on her own solved sheet or annotated slide. Worked by the drill: the drill's own working, where her sheet gives no worked value.

a

Her key

Answer: 121.17 mL/min

WHAT THE STEM GIVES

dose	500 mg oral every 12 hours for 10 days	*not needed for CrCl*
age	42 year old	140 − 42 = 98 on top
sex	male	IBW = 50 + 2.3 × inches over 5 ft; no 0.85 factor
height	178 cm	$\frac{178}{2.54}$ = 70.08 in, 10 in over 5 ft, so IBW = 73 kg
weight	73 kg	*not needed: IBW is used, and here it also comes to 73 kg*
S_{Cr}	0.82 mg/dL	the denominator, 72 × 0.82 = 59.04
F	95%	*not needed*
t½	1.3 hours	*not needed for CrCl*
V_D	0.23 L/kg	*not needed*
f_e	60%	*not needed*

THE WORKING, ONE LINE AT A TIME

UNIT

178 cm ÷ 2.54 = 70.08 in, taken as 10 in over 5 ft

Height in centimetres is divided by 2.54 cm per inch: $\frac{178\ \text{cm}}{2.54\ \text{cm/in}}$ = 70.08 in, the centimetres cancelling. The ideal body weight equation counts inches above 60 in (5 ft), so 70.08 − 60 = 10.08, rounded to a whole 10 in before the IBW is computed, as her keys do.

ALGEBRA

IBW = 50 + 2.3(10) = 73 kg

The male form of the ideal body weight equation starts at 50 kg at 5 ft and adds 2.3 kg for each inch over 5 ft: 50 + 2.3(10) = 50 + 23 = 73 kg. KL weighs 73 kg, so here IBW and actual weight coincide; the equation uses IBW either way.

SETUP

$$\text{CrCl} = \frac{(140 - 42)(73\ \text{kg})}{(72)(0.82)} = 121.17\ \text{mL/min}$$

This is the Cockcroft-Gault equation; the patient is male, so the 0.85 factor for a female patient is not applied.

> **DOES THE NUMBER MAKE SENSE?**
>
> A 42-year-old man with S_{Cr} 0.82 mg/dL has normal kidneys, so a CrCl near 120 mL/min fits and 121.17 mL/min does; the stem itself puts renal impairment between 30 and 120 mL/min.

With 10.08 inches instead of 10, IBW is 73.18 kg and CrCl 121.47 mL/min; both are accepted.

---

b

Her key

Answer: 149 mL/min

WHAT THE STEM GIVES

weight	73 kg	$V_D = 0.23 \times 73 = 16.79$ L
V_D	0.23 L/kg	times 73 kg gives 16.79 L, multiplied by k
t½	1.3 hours	normal renal function applies: $k = \frac{0.693}{1.3} = 0.533\ hr^{-1}$
t½ impaired	5.2 hours	*not needed: KL has normal renal function*
t½ failure	5.9 hours	*not needed*
f_e	60%	*not needed: the total is asked*
F	95%	*not needed*
dose	500 mg	*not needed*

THE WORKING, ONE LINE AT A TIME

SETUP

$$Cl_T = (0.23\ L/kg)(73\ kg)(\frac{0.693}{1.3\ hr}) = 8.95\ L/hr$$

Total clearance $Cl_T = V_D \times k$, with $k = \frac{0.693}{t½}$. V_D, the volume of distribution, is 0.23 L/kg × 73 kg = 16.79 L; KL has a CrCl of 121.17 mL/min, normal renal function, so the 1.3 hr half-life applies and k = 0.533 hr^{-1}; (16.79 L)(0.533 hr^{-1}) = 8.95 L/hr.

UNIT

$$(8.95\ L/hr)(\frac{1000\ mL}{L})(\frac{1\ hr}{60\ min}) = 149\ mL/min$$

Converting to mL/min puts total clearance in the same unit as creatinine clearance, so the two can be compared.

> **DOES THE NUMBER MAKE SENSE?**
> A 1.3 hour half-life clears about half of the 0.23 × 73 L volume each hour, which is near 8.95 L/hr; one L/hr is about 17 mL/min, so about 150 mL/min fits, and 149 mL/min does.

---

c

Her key

Answer: 89.5 mL/min

WHAT THE STEM GIVES

weight	73 kg	$V_D = 0.23 \times 73 = 16.79$ L
V_D	0.23 L/kg	with k gave Cl_T = 149 mL/min in part b
t½	1.3 hours	gives $k = \frac{0.693}{1.3} = 0.533\ hr^{-1}$ for Cl_T
f_e	60%	f_e = 0.6 multiplies Cl_T = 149 mL/min from part b
F	95%	*not needed*

dose	500 mg	*not needed*

THE WORKING, ONE LINE AT A TIME

SETUP

$Cl_R = f_e \times Cl_T = (0.6)(149\ mL/min)$

About 60% of the drug is excreted unchanged in urine, so $f_e = 0.6$ and renal clearance is 0.6 of total clearance.

ROUND

$Cl_R = 89.5\ mL/min$

(0.6)(149 mL/min) = 89.4 mL/min with the rounded total; the unrounded total clearance is 149.17 mL/min, and 0.6 of it is 89.50 mL/min, the value her key reports. f_e has no units, so Cl_R keeps mL/min, and both results fall within the tolerance.

> **DOES THE NUMBER MAKE SENSE?**
> Six tenths of 149 mL/min is about 90 mL/min, so 89.5 mL/min; Cl_R must be below the total clearance, 149 mL/min.

(0.6)(149) gives 89.4; her key prints 89.5 from the unrounded 149.17. Either is accepted.

---

d

Her key

Answer: 59.5 mL/min

WHAT THE STEM GIVES

weight	73 kg	$V_D = 0.23 \times 73 = 16.79$ L
V_D	0.23 L/kg	with k gave $Cl_T = 149$ mL/min in part b
t½	1.3 hours	gives $k = \frac{0.693}{1.3} = 0.533\ hr^{-1}$ for Cl_T
f_e	60%	$Cl_R = 89.5$ mL/min from part c, subtracted from Cl_T
F	95%	*not needed*
dose	500 mg	*not needed*

THE WORKING, ONE LINE AT A TIME

SETUP

$Cl_H = Cl_T - Cl_R = (149 - 89.5)\ mL/min = 59.5\ mL/min$

Total clearance Cl_T is the sum of the clearances by each route, $Cl_T = Cl_R + Cl_H$, so the hepatic clearance Cl_H is what remains after the renal clearance Cl_R is subtracted: (149 − 89.5) mL/min = 59.5 mL/min, the 40% of the drug not excreted unchanged.

> **DOES THE NUMBER MAKE SENSE?**
> Cl_H is what is left of 149 mL/min after the renal 89.5 mL/min: 149 − 89.5 = 59.5 mL/min, below Cl_T and below Cl_R because more than half is renal.

---

e

Her key

Answer: 35.49 mL/min

WHAT THE STEM GIVES

age	42 year old	140 − 42 = 98 on top
sex	male	IBW = 50 + 2.3 × inches over 5 ft; no 0.85 factor
height	178 cm	10 in over 5 ft, so IBW = 73 kg, as in part a
weight	73 kg	*not needed: IBW is used*
S_{Cr}	0.82 mg/dL	*not needed: replaced by the new value*
S_{Cr} new	2.8 mg/dL	the new denominator, 72 × 2.8 = 201.6

THE WORKING, ONE LINE AT A TIME

SETUP

$$\text{CrCl} = \frac{(140 - 42)(73\ \text{kg})}{(72)(2.8)} = 35.49\ \text{mL/min}$$

Only the serum creatinine changes; it sits in the denominator, so a higher S_{Cr} gives a lower creatinine clearance.

> **DOES THE NUMBER MAKE SENSE?**
> S_{Cr} rose from 0.82 to 2.8 mg/dL, more than three times, so CrCl falls to less than a third of the part a estimate; 35.49 mL/min lies in the 30 to 120 mL/min band the stem calls renal impairment.

f

Her key

Answer: 37.29 mL/min

WHAT THE STEM GIVES

weight	73 kg	V_D = 0.23 × 73 = 16.79 L, unchanged
V_D	0.23 L/kg	unchanged; times 73 kg gives 16.79 L
t½ impaired	5.2 hours	applies now that CrCl lies between 30 and 120: k = $\frac{0.693}{5.2}$
t½	1.3 hours	*not needed: normal function no longer applies*
t½ failure	5.9 hours	*not needed: CrCl is above 30*
f_e	60%	*not needed: the total is asked*

THE WORKING, ONE LINE AT A TIME

SETUP

35.49 mL/min lies between 30 and 120, so t½ = 5.2 hr

The stem gives a half-life for each CrCl band; 35.49 mL/min lies between 30 and 120, so the 5.2 hr value applies.

ALGEBRA

$$Cl_T = (0.23\ L/kg)(73\ kg)(\frac{0.693}{5.2\ hr}) = 2.2376\ L/hr$$

The volume of distribution is unchanged and the half-life is now 5.2 hr, so k and the clearance are both smaller.

UNIT

$$(2.2376\ L/hr)(\frac{1000\ mL}{L})(\frac{1\ hr}{60\ min}) = 37.29\ mL/min$$

L/hr to mL/min: × 1000 turns litres into millilitres, 2.2376 × 1000 = 2237.6 mL/hr, and dividing by 60 min/hr turns hours into minutes, $\frac{2237.6\ mL/hr}{60\ min/hr}$ = 37.29 mL/min, her key value. mL/min puts clearance in the unit of CrCl.

> **DOES THE NUMBER MAKE SENSE?**
> One L/hr is about 17 mL/min, so 2.2376 L/hr is about 2 × 17 plus a little, near 37.29 mL/min; the half-life is four times the 1.3 hours, so this is a quarter of the part b clearance.

Rounding Cl_T to 2.24 L/hr first gives 37.33 mL/min; carrying 2.2376 L/hr gives her key value, 37.29. Either is accepted.

---

g

Her key

Answer: 22.38 mL/min

WHAT THE STEM GIVES

V_D	0.23 L/kg	with the new k gave Cl_T = 37.29 mL/min in part f
t½ impaired	5.2 hours	gives k = $\frac{0.693}{5.2}$ hr^{-1} for the new Cl_T
f_e	60%	f_e = 0.6 multiplies the new Cl_T, 37.29 mL/min from part f

THE WORKING, ONE LINE AT A TIME

SETUP

$$Cl_R = (0.6)(37.29\ mL/min) = 22.38\ mL/min$$

Renal clearance $Cl_R = f_e \times Cl_T$. The fraction excreted unchanged, f_e = 0.6, describes how the drug is handled and does not change with the serum creatinine, so Cl_R is still 0.6 of the total, now the lower 37.29 mL/min: (0.6)(37.29) = 22.37, which her key rounds to 22.38 from the unrounded total.

> **DOES THE NUMBER MAKE SENSE?**
> Six tenths of 37.29 mL/min is a little over 22, so 22.38 mL/min; Cl_R stays below the total, 37.29 mL/min.

---

h

Her key

Answer: 14.91 mL/min

WHAT THE STEM GIVES

V_D	0.23 L/kg	with the new k gave Cl_T = 37.29 mL/min in part f
t½ impaired	5.2 hours	gives k = $\frac{0.693}{5.2}$ hr^{-1} for the new Cl_T
f_e	60%	Cl_R = 22.38 mL/min from part g, subtracted from Cl_T

THE WORKING, ONE LINE AT A TIME

SETUP

$$Cl_H = (37.29 - 22.38)\ mL/min = 14.91\ mL/min$$

Total clearance Cl_T is the sum of the renal and hepatic clearances, $Cl_T = Cl_R + Cl_H$, so $Cl_H = Cl_T - Cl_R = 37.29 - 22.38 = 14.91$ mL/min. Both terms are in mL/min, so the difference is too; it is 0.4 of the total, the share not excreted unchanged.

> **DOES THE NUMBER MAKE SENSE?**
> 37.29 − 22.38 = 14.91 mL/min, the part of the total not handled by the kidney; it is below Cl_T, 37.29, and below Cl_R, 22.38, because more than half is renal.